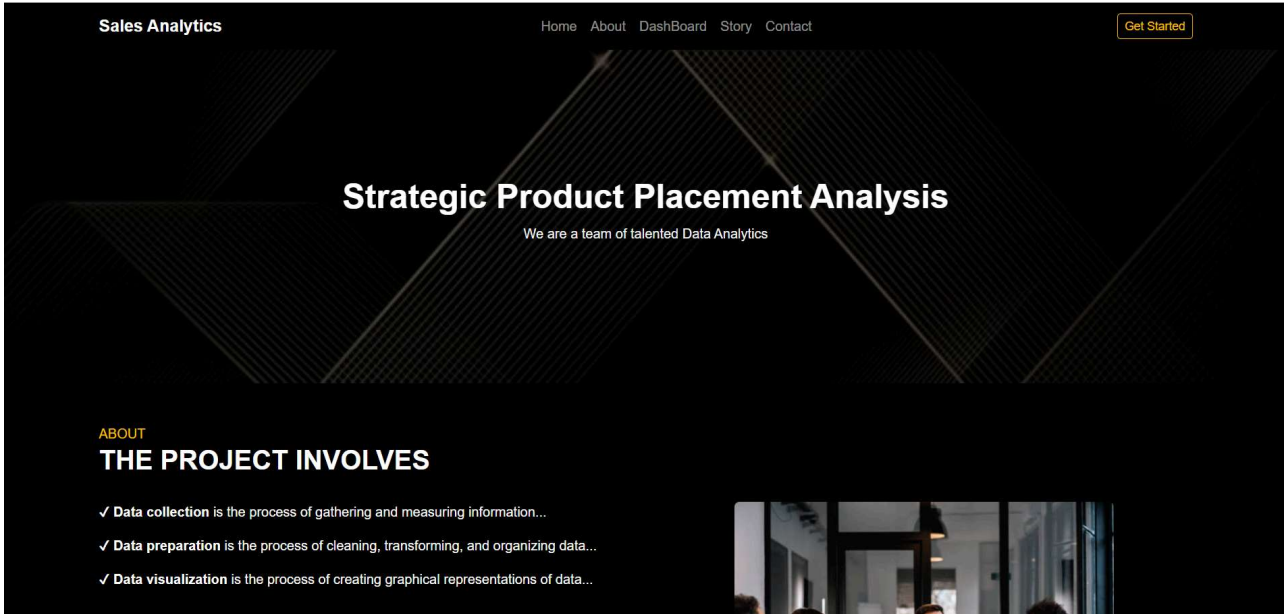


# Project Design Phase-II

## Technology Stack (Architecture & Stack)

|         |              |               |                                             |
|---------|--------------|---------------|---------------------------------------------|
| Date    | 30 June 2026 | Maximum Marks | 4 Marks                                     |
| Team ID |              | Project Name  | Strategic Product Placement Analysis System |



## Technical Architecture:

The system architecture relies on a decoupled microservices approach, processing high-volume retail POS data and rendering spatial heatmaps on a web frontend.

| S.No | Component           | Description Technology                                                                                                                                                        |
|------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | User Interface      | Interactive web dashboard for retail analysts and mobile view for store managers to access floor plans and placement heatmaps.<br>React.js, HTML5, CSS3, D3.js (for Heatmaps) |
| 2.   | Application Logic-1 | Backend API routing, user authentication, and system orchestration.<br>Node.js (Express) or Python (FastAPI)                                                                  |

|    |                               |                                                                                                                |                                  |
|----|-------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------|
| 3. | Application Logic-2           | Spatial mapping engine to parse floor plan images into a digital coordinate grid.                              | Python, OpenCV, Pandas           |
| 4. | Database                      | Core relational database storing user profiles, store metadata, and access permissions.                        | PostgreSQL                       |
| 5. | Cloud Database                | Scalable database for storing large volumes of POS coordinates and heatmap nodes.                              | MongoDB (NoSQL)                  |
| 6. | File Storage                  | Cloud storage for raw store floor plan files (PNG/JPG) and generated strategy reports (PDF).                   | AWS S3 / Cloud Object Storage    |
| 7. | Machine Learning Model        | Predictive analytics model to calculate high traffic zones and impulse-buy hotspots based on historical sales. | Python, Scikit-Learn, TensorFlow |
| 8. | Infrastructure (Server/Cloud) | Containerized deployment of microservices on a scalable cloud infrastructure.                                  | Docker, Kubernetes, AWS/ GCP     |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                                                 | Technology                                   |
|------|--------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1.   | Open-Source Frameworks   | Utilization of open-source libraries for frontend visualization and backend data parsing.                   | React, D3.js, OpenCV, Pandas                 |
| 2.   | Security Implementations | Data encryption in transit, secure token-based authentication, and hashed passwords.                        | JWT (JSON Web Tokens), bcrypt, HTTPS/TLS 1.2 |
| 3.   | Scalable Architecture    | Containerized microservices enabling independent scaling of the ML prediction engine vs. the web dashboard. | Docker, Kubernetes Orchestration             |

|    |              |                                                                                                                   |                                         |
|----|--------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 4. | Availability | Distribution of traffic across multiple server instances to prevent downtime during peak retail analysis periods. | Cloud Load Balancers<br>(e.g., AWS ALB) |
| 5. | Performance  | In-memory caching of frequently accessed heatmap coordinates to ensure sub-3-second load times.                   | Redis Caching                           |